

PRATHOSH H

BIOMEDICAL ENGINEERING

✉ hprathosh1@gmail.com 📍 Madurai, India, 625002

☎ +919787590195

🌐 prathosh-h-ha27012006

CAREER OBJECTIVE

Biomedical Engineering student seeking an entry-level opportunity in medical device manufacturing, biomedical instrumentation, and quality engineering. Interested in applying technical knowledge to support safe, reliable, and innovative healthcare solutions.

EDUCATION

2023–2027

B.E. Biomedical Engineering

Sethu Institute of Technology

CGPA: 8.799/10

2023

H.S.E.

Sairan Matric Higher Secondary School

Percentage: 74.41%

ACADEMIC PROJECTS

- **EYEZEN – Intelligent Eye Fatigue Regulation** Third-Year
Through Adaptive Illumination Control
 - Developed an adaptive lighting system to reduce digital eye strain using BH1750, PIR and IR sensors.
 - Designed automatic illumination control to maintain approximately 400 lux based on environmental conditions and user presence.
 - Proposed a scalable solution for classrooms, offices, computer labs and smart workspaces.

- **TMJ Pain Reliever – Wearable Therapeutic Device** Second-Year
 - Developed a wearable device concept for non-invasive TMJ pain relief.
 - Integrated electronic components for controlled therapeutic stimulation.
 - Focused on compact design, user comfort and safe wearable operation.

AREAS OF INTEREST

- Biomedical Instrumentation & Electronic Systems
- Medical Devices & Diagnostic Equipment
- Electrical & Electronic Measurements
- Sensors, Signal Conditioning & Data Acquisition
- Biomedical Signal Processing
- Medical Device Manufacturing & Quality
- Automation Industry

SKILLS

- Programming: C, C++, Python
- Software: MATLAB, Scilab,
- Electronics & Instrumentation: Basic Electronic Measurements, Sensors, Signal Conditioning, Data Acquisition
- Biomedical: Biomedical Instrumentation, Biomedical Signal Processing, Medical Devices
- Embedded Systems: Arduino, Raspberry Pi, ESP32

ACHIEVEMENTS

- 2nd Prize – Project Expo, Kalasalingam University
- Participant – EYEQ Innovate Hackathon 2.0
- Participated in technical symposiums and biomedical innovation events
- Participant in Markers Arena 2.0 24 hours Hackthon [Hardware]

INTERNSHIP

Vadamalayan Hospitals Pvt. Ltd.

Biomedical Engineering Intern

- Assisted with biomedical equipment inspection and basic maintenance.
- Gained exposure to preventive maintenance and troubleshooting.

Saravana Multi-Speciality Hospital Pvt. Ltd.

Biomedical Engineering Intern

- Observed biomedical equipment operation and maintenance practices.
- Gained exposure to equipment safety and documentation.

CERTIFICATIONS

- C Programming
- Healthcare Entrepreneurship
- Biomedical Equipment product based training

LANGUAGES

- Tamil – Native
- English – Professional Working Proficiency
- Hindi -- Basic / Beginner